

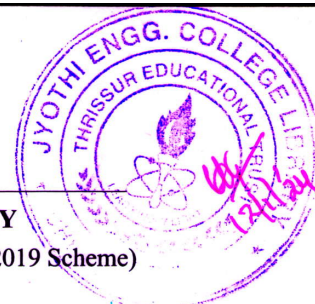
Reg No.: \_\_\_\_\_

Name: \_\_\_\_\_

1100EET305122105

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S5 (R, S) / S5 (PT) (R, S) Examination December 2023 (2019 Scheme)



Course Code: EET 305  
Course Name: SIGNALS AND SYSTEMS

Max. Marks: 100

Duration: 3 Hours

**PART A**

*(Answer all questions; each question carries 3 marks)*

Marks

- 1 Define unit step signal. Sketch the signal  $x(t) = -u(-t - 1)$  (3)
- 2 Check whether the system,  $y(t) = e^{-2x(t)}$  is linear. (3)
- 3 Find the Fourier transform of  $x(t) = e^{-t} u(t)$  (3)
- 4 Derive the elements in electrical system analogous to mechanical translational system with respect to force-voltage analogy. (3)
- 5 Determine the unit impulse response for the system with transfer function  $H(s) = \frac{10}{s+100}$  (3)
- 6 List down any three properties of positive real functions. (3)
- 7 Define sampling and state sampling theorem. (3)
- 8 List down any three properties of z-transform. (3)
- 9 Using bilinear transformation method, check the stability of a discrete system having characteristic equation as:  $z^2 + 0.2z + 0.1$ . (3)
- 10 Find the impulse response of a discrete time system having transfer function as:  $H(z) = \frac{z}{z-2}$  (3)

**PART B**

*(Answer one full question from each module, each question carries 14 marks)*

**Module -1**

- 11 a) Differentiate between energy and power signals. Determine whether the signal,  $x(t) = e^{-t}u(t)$  is an energy signal or power signal. (7)
- b) Find the convolution integral of the above signal  $x(t) = e^{-t}u(t)$  with another signal  $h(t) = e^{-3t}u(t)$  (7)

- 12 a) Test whether the following signals are periodic or not. If periodic, find the fundamental period (5)

$$x(t) = 5 \sin 24\pi t + 7 \sin 36\pi t$$

- b) With suitable examples, differentiate between (9)
- Linear and non-linear systems
  - Variant and Time-variant systems
  - Stable and unstable systems

### Module -2

- 13 a) Find the exponential Fourier Series for the signal shown in Fig.1. (9)  
Given  $T = 2\pi$  and  $V = 10$ .

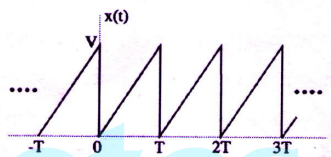


Fig.1

- b) State and prove the time shifting property of Fourier Transform. (5)
- 14 a) Derive the transfer function  $\frac{X(s)}{F(s)}$  of the mechanical system shown in Fig.2 (9)

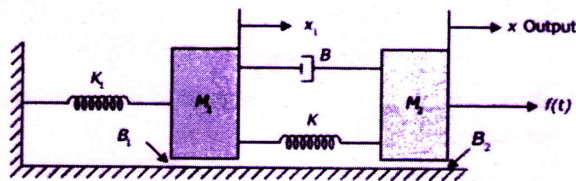


Fig.2

- b) A certain system is described by the following differential equation (5)

$$\frac{d^2 y(t)}{dt^2} + 7 \frac{dy(t)}{dt} + 12y(t) = x(t)$$

Use Laplace transform to determine the response of the system to unit step input applied at  $t = 0$ .

Assume the initial conditions on the system are  $y(0) = -2$ ,  $\frac{dy(0)}{dt} = 0$

## Module -3

- 15 a) Determine the overall transfer function using block diagram reduction for system (5) shown in Fig.3.

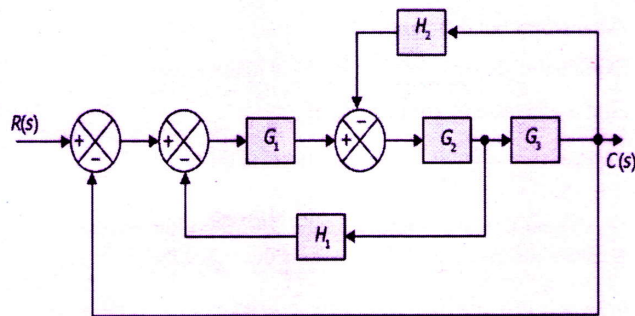


Fig.3

- b) For the closed loop system shown in shown in Fig.4, find the transfer function (9) and thus check the stability of the system if  $G(s) = \frac{s}{s+2}$  and  $H(s) = \frac{s}{s+1}$  using Routh Hurwitz criterion.

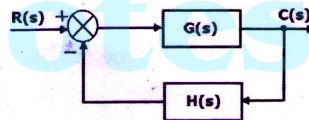


Fig.4

- 16 a) Determine the transfer function of the system shown in Fig.5 represented by the (9) signal flow graph using Mason's Gain Formula.

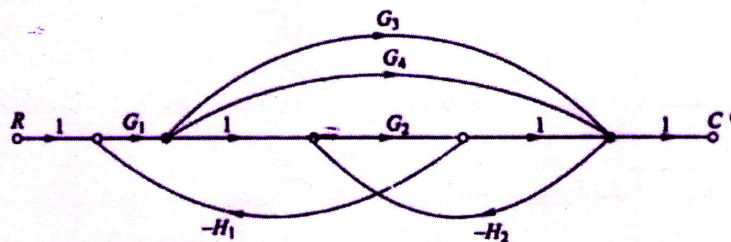


Fig.5

- b) Check the stability of the system represented by the following characteristic (5) equation using Routh stability criterion:  $s^3 + 3s^2 + 4s + 2$



## Module -4

- 17 a) Determine the convolution sum of the 2 sequences (5)  
 $x(n) = \{1, 2, 3, 4\}$  and  $h(n) = \{1, 2, 1\}$
- b) Show that the sequences  $x(n) = a^n u(n)$  and  $y(n) = -a^n u(-n-1)$  have same (9)  
 Z-transform but different ROC.
- 18 a) Explain the aliasing effect in sampled data systems. (5)
- b) Find the inverse of z-transform of: (9)  
 $H(z) = \frac{3z^{-1}}{(1-z^{-1})(1-2z^{-1})}$ . Given ROC is  $|z| > 2$

## Module -5

- 19 a) A causal discrete time LTI system is described by (9)  
 $y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n)$   
 where  $x(n)$  and  $y(n)$  are the input and output of the system, respectively.  
 Determine the transfer function  $H(z)$  of the system. Also find the step response of the system.
- b) Determine the Fourier series of the sequence  $x(n) = \cos 0.2\pi n$  (5)
- 20 a) Obtain the Direct form-I realization for the system described by the difference (9)  
 equation  
 $y(n) + \frac{1}{4}y(n-1) - \frac{1}{8}y(n-2) = x(n) - \frac{7}{4}x(n-1) - \frac{1}{2}x(n-2)$
- b) A system is described by the following characteristic equation:  $5z^4 + 2z^3 + (5)$   
 $Kz^2 + 3z + 1$  Using Jury's test, find the value of  $K$  that will result in a stable system.

\*\*\*